Bisection Method

1) $F(x) = x^3 - x - 1 = 0$. interval $[1, 2]$

$F(1) = 1^3 - 1 - 1$
$= -1$

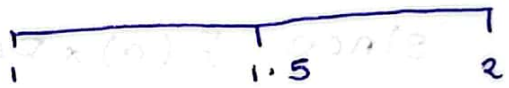
$F(2) = 2^3 - 2 - 1$
$= 6 - 1$
$= 5$

$F(1) \times F(5) < 0$

So we can find the root

Step 1 $\Rightarrow$ $c_1 = \frac{1+2}{2} = 1.5$

Now calculate $F(c_1)$



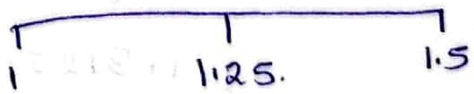
$F(c_1) = 1.5^3 - 1.5 - 1$
$= 0.875$

since $F(1) \times F(c_1) < 0$
the root lies in $(1, 1.5)$

step 2 :-
Now update the interval.

$a = 1$
$b = 1.5$

$c_2 = \frac{1+1.5}{2}$
$= 1.25$



then, calculate $F(c_2)$

$F(c_2) = 1.25^3 - 1.25 - 1$
$= -0.297.$

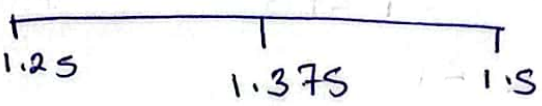
since $F(1) \times F(c_2) > 0$. the root lies in $(1.25, 1.5)$.

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Step 3:-
Now update the interval

$a = 1.25$
$b = 1.5$

$$C_3 = \frac{1.25+1.5}{2}$$



$$= 1.375$$

$$F(C_3) = 1.375^3 - 1.375 - 1$$

$$= 0.2246$$

$$= 0.225$$

since $F(a) \times F(C_3) < 0$

$- \times + < 0$

The root lies in $(1.25, 1.375)$

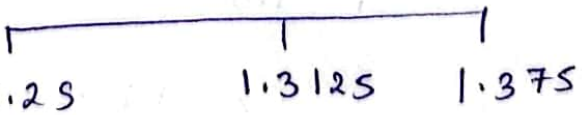
Step 4:-

Now update the interval

$a = 1.25$

$b = 1.375$

$$C_4 = \frac{1.25+1.375}{2}$$



$$= 1.3125$$

Now, $F(C_4) = 1.3125^3 - 1.3125 - 1$

$$= -0.052$$

since $F(a) \times F(C_4) > 0$

$- \times - > 0$

The root lies in $(1.3125, 1.375)$

Steps :-

Now, update the interval

$a = 1.3125$

$b = 1.375$.

$$c_5 = \frac{1.3125 + 1.375}{2}$$

$$= 1.34375.$$

So Five iteration roots are

$c_1 = 1.5$	(1, 2)
$c_2 = 1.25$	(1, 1.5)
$c_3 = 1.375$	(1.25, 1.5)
$c_4 = 1.3125$	(1.25, 1.375)
$c_4 = 1.34375$	(1.3125, 1.375)

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2) Fixed point iteration method

Rewrite in the Form $x = g(x)$

we need to isolate x on one side

$$x^3 - x - 1 = 0$$

$$x = \sqrt[3]{x+1}$$

use Fixed point iteration Formula.

$$x_{n+1} = g(x_n).$$

Initial root $x_0 = 1.5$

it.

step 1 :- $x_1 = g(x_0)$

$= \sqrt[3]{1.5+1}$

$= 1.357$

step 2 :- $x_2 = g(x_1)$

$= \sqrt[3]{1.357+1}$

$= 1.331$

step 3 :- $x_3 = g(x_2)$

$= \sqrt[3]{1.331+1}$

$= 1.326$

step 4 :- $x_4 = g(x_3)$

$= \sqrt[3]{1.326+1}$

$= 1.325$

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Step 5 :-

$$x_5 = g(x_4)$$

$$= \sqrt[3]{1.325 + 1}$$

$$= 1.325.$$

$x_0 = 1.5$

$x_1 = 1.357$

$x_2 = 1.331$

$x_3 = 1.326$

$x_4 = 1.325.$

$x_5 = 1.325.$

3) Secant Method.

$x_0 = 1$

$x_1 = 2$.

$F(x) = x^3 - x - 1$

$F(1) = -1$

$F(2) = 5$

$F(1) \times F(2) < 0$

so the root lies in $[1, 2]$

Use Secant Formula.

$$x_{n+1} = x_n - F(x_n) \cdot \frac{(x_n - x_{n-1})}{F(x_n) - F(x_{n-1})}$$

Iteration 1 :-

$$x_2 = x_1 - F(x_1) \frac{(x_1 - x_0)}{F(x_1) - F(x_0)}$$

$$= 2 - 5 \frac{(2-1)}{(5+1)}$$

$$= 2 - \frac{5}{6} = \frac{7}{6}$$

$= 2.0833$

$= 1.167$

Now update Interval

$x_0 = 2$ $\quad F(x_0) = 5$

$x_1 = 1.167$. $\quad F(x_1) = -0.578$

Iteration 2.

$$x_2 = x_1 - F(x_1) \cdot \frac{(x_1 - x_0)}{F(x_1) - F(x_0)}$$

$$= 1.167 - (-0.578) \frac{(1.167 - 2)}{-0.578 - 5}$$

$$= 1.167 - \frac{0.481 \times 500}{-2789}$$

$$= 1.167 + 0.086$$

$$= 1.253$$

update interval

$x_0 = 1.167$ $\quad F(x_0) = -0.578$

$x_1 = 1.253$ $\quad F(x_1) = -0.286$

Iteration 3 :-

$$x_2 = 1.253 - (-0.286) \frac{(1.253 - 1.167)}{-0.286 + 0.578}$$

$$= 1.253 + \frac{(-0.025)}{0.292}$$

$$= 1.253 + 0.086$$

$$= 1.337$$

Now, update interval

$x_0 = 1.253$ ~~$F(x_0) = -0.578$~~

$F(x_0) = -0.286$

$x_1 = 1.337$

$F(x_1) = 0.053$

Iteration 4 :-

$$x_2 = 1.337 - 0.053 \frac{(1.337 - 1.253)}{0.053 + 0.286}$$

$$= 1.337 - 0.013$$

$$= 1.324$$

Now, update interval

$x_0 = 1.337$ $F(x_0) = 0.053$

$x_1 = 1.324$ $F(x_1) = 0.003$

Iteration 5 :-

$$x_2 = 1.324 - 0.003 \frac{(1.324 - 1.337)}{0.003 - 0.053}$$

$$= 1.324 + \frac{3.9 \times 10^{-5}}{-0.05}$$

$$= 1.324 - 0.0008$$

$$= 1.323$$

∴

root $\rightarrow x_5 \rightarrow$ Iteration 5

1.323

4) False position method.

Initial setup:

$a = 1 \quad F(1) = -1$

$b = 2 \quad F(2) = 5.$

check $F(a) \times F(b) < 0.$

The root lies in $[1, 2]$

Iteration 1:

$$x_1 = b - \frac{F(b)(b-a)}{F(b) - F(a)}$$

$$x_1 = 2 - \frac{5(2-1)}{5+1}$$

$$x_1 = 2 - \frac{5}{6}$$

$$= \frac{7}{6}$$

$$= 1.667$$

$F(x_1) = -0.578$

$F(a) \times F(x_1) > 0$

So update.

$a = x_1 \Rightarrow a = 1.667.$

$b = 2$

Iteration 2

$$x_2 = 2 - 5\,\frac{(2-1.167)}{(5+0.578)}$$

$$= 2 - \frac{5 \times 0.833}{5.578}$$

$$= 2 - 0.747.$$

$$= 1.253$$

$$F(x_2) = 1.253^3 - 1.253 - 1$$

$$= -0.286$$

$$F(a) \times F(x_2) > 0.$$

So.

update interval

$a = x_2. \Rightarrow a = 1.253$

$b = 2.$

iteration. 3

$$x_3 = 2 - 5\,\frac{(2-1.253)}{5+0.286.}$$

$$= 2 - 0.707$$

$$= 1.293.$$

$$F(x_3) = -0.131$$

$$F(a) \times F(x_3) = -x- > 0.$$

So.

update interval

$a = 1.293$

$b = 2$

Iteration 4 :-

$$x_4 = 2 - \frac{5(2-1.293)}{5+0.131}$$

$$= 2 - 0.689$$

$$= 1.311$$

$$F(x_4) = -0.058$$

$$F(a) \times F(x_4) = -x- > 0$$

so

update interval

$a = 1.311$

$b = 2$

Iteration 5 :-

$$x_5 = 2 - \frac{5(2-1.311)}{5+0.058}$$

$$= 2 - 0.681$$

$$= 1.319$$

5) Newton Raphson method

Formula

$$x_{n+1} = x_n - \frac{F(x_n)}{F'(x_n)}$$

$F(x) = x^3 - x - 1$

$F'(x) = 3x^2 - 1$

Iteration 1.

Initial $x_0 = 1.5$

$F(1.5) = 0.875$

$F'(1.5) = 5.75$

$$x_1 = 1.5 - \frac{0.875}{5.75}$$

$$= 1.348$$

Iteration 2

$x_0 = 1.348$

$F(x_0) = F(1.348) = 0.101$

$F'(1.348) = 4.451$

$$x_2 = 1.348 - \frac{0.101}{4.451}$$

$$= 1.325$$

Iteration 3

$F(1.325) = 0.0012$

$F'(1.325) = 4.267$

$$x_3 = 1.325 - \frac{0.0012}{4.267}$$

$= 1.325$

$= 1.325.$

So. our root is. 1.325.

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